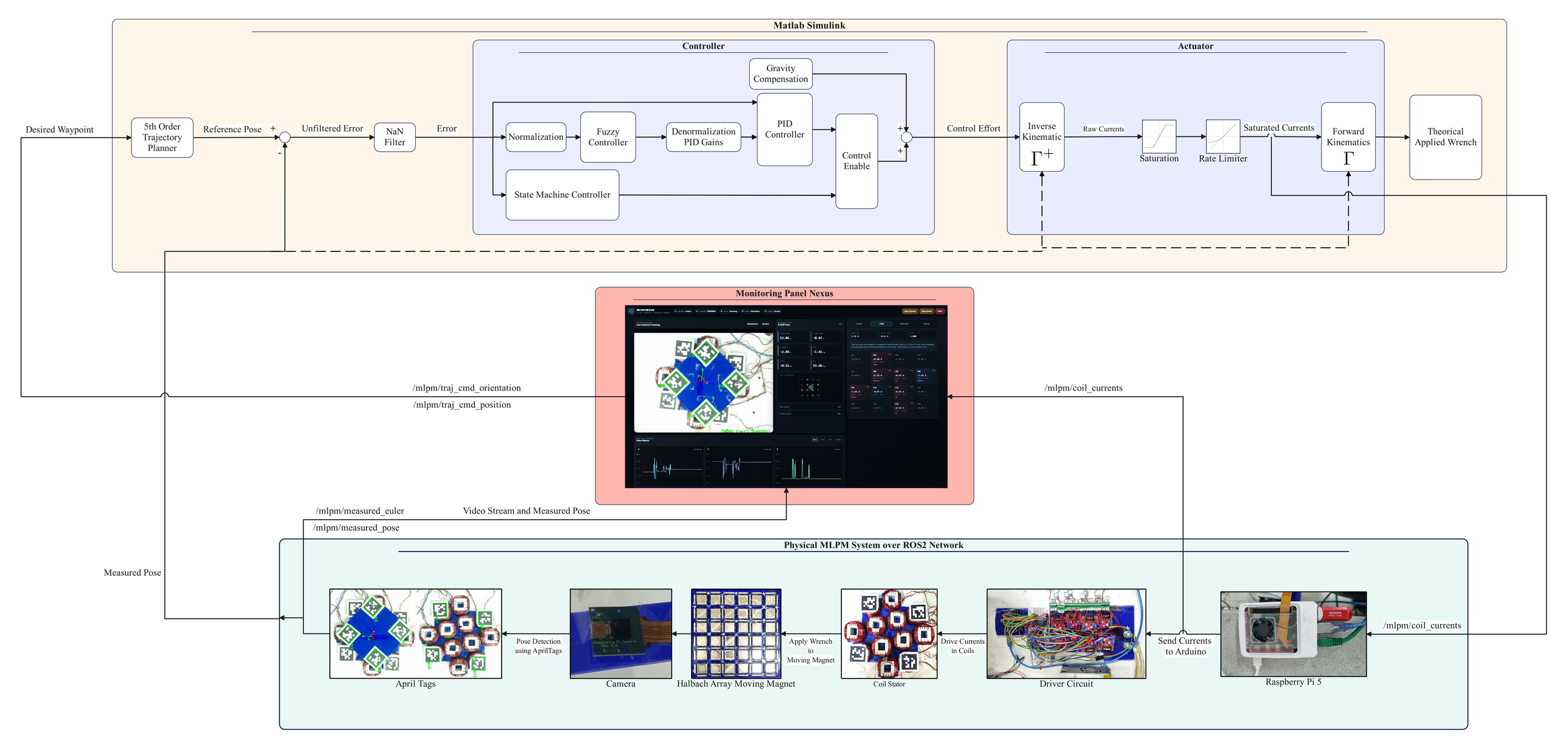

Matlab Simulink
Controller
Actuator
Desired Waypoint
5th Order Trajectory Planner
Reference Pose
+
-
Unfiltered Error
NaN Filter
Error
Normalization
Fuzzy Controller
Denormalization PID Gains
Gravity Compensation
PID Controller
State Machine Controller
Control Enable
+
+
Control Effort
Inverse Kinematic
Γ+
Raw Currents
Saturation
Rate Limiter
Saturated Currents
Forward Kinematics
Γ
Theorical Applied Wrench
Monitoring Panel Nexus
/mlpm/traj_cmd_orientation
/mlpm/traj_cmd_position
/mlpm/coil_currents
/mlpm/measured_euler
/mlpm/measured_pose
Video Stream and Measured Pose
Measured Pose
Physical MLPM System over ROS2 Network
April Tags
Pose Detection using AprilTags
Camera
Halbach Array Moving Magnet
Apply Wrench to Moving Magnet
Coil Stator
Drive Currents in Coils
Driver Circuit
Send Currents to Arduino
Raspberry Pi 5
/mlpm/coil_currents